VoIP – [Write up]

Your close friend James recently received a suspicious phone call from someone claiming to be his bank. The caller asked for sensitive information, making James uneasy. Suspecting a potential Vishing (Voice Phishing) attack, you decide to investigate by capturing and analyzing the VoIP traffic.

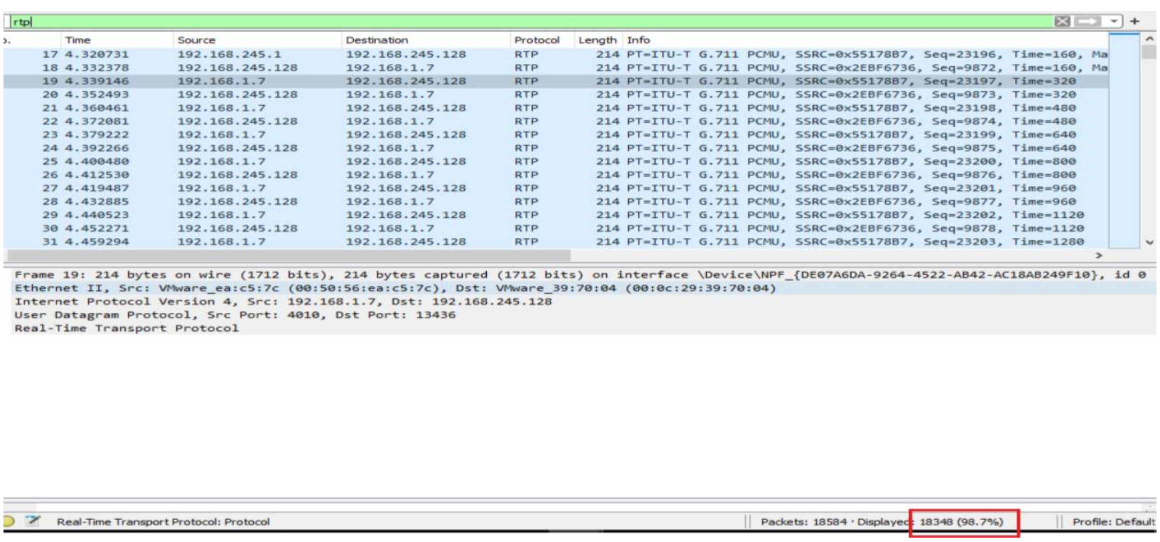
Note: To listen to the VoIP call, you must access the machine via RDP.

Technical Overview of VoIP Protocols

Voice over Internet Protocol (VoIP) relies on two primary open-standard protocols working in tandem to enable real-time communication over IP networks: **SIP** (Session Initiation Protocol) and **RTP** (Real-time Transport Protocol). SIP operates as the signaling and control layer, responsible for establishing, managing, authenticating, and terminating connections between endpoints, while encapsulating metadata such as caller identities and timestamps. Once a session is initialized, RTP takes over as the media transport layer, converting voice signals into digital payloads and transmitting them rapidly across the network using UDP to ensure low-latency, real-time audio delivery.

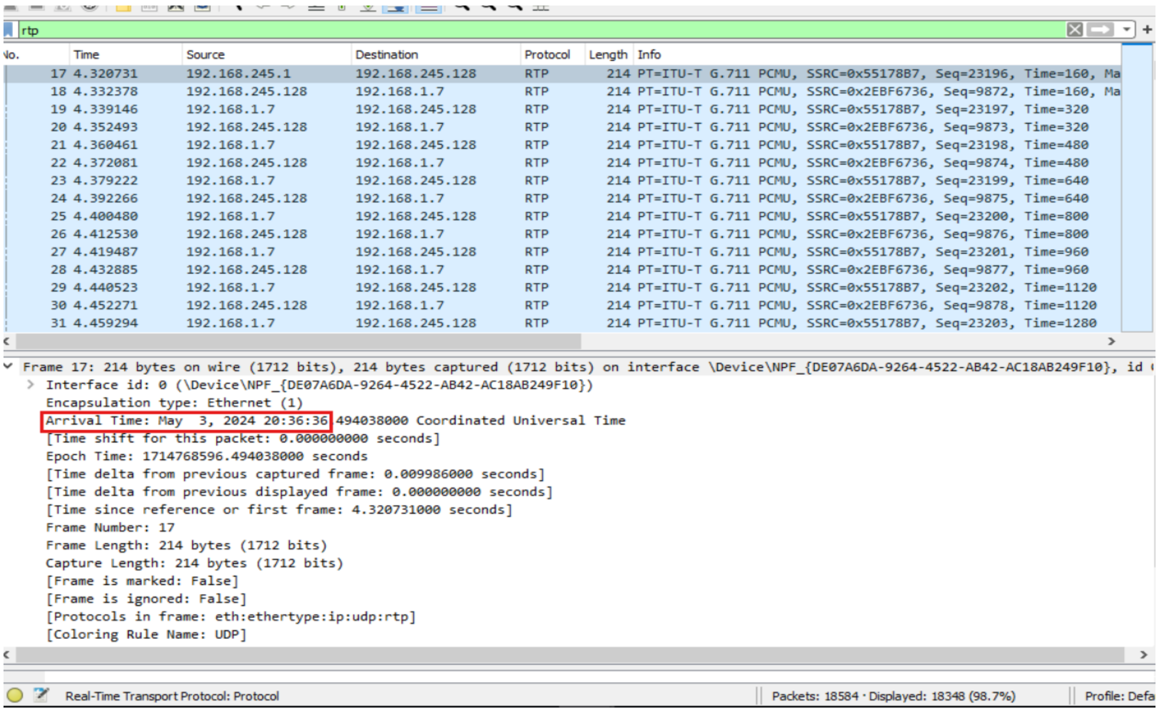
| Start Investigation

Q1: How many RTP packets were in the traffic?



For the first question, I searched for "rtp" and the number of packets appeared.

Q2: When did the fake call with James start?

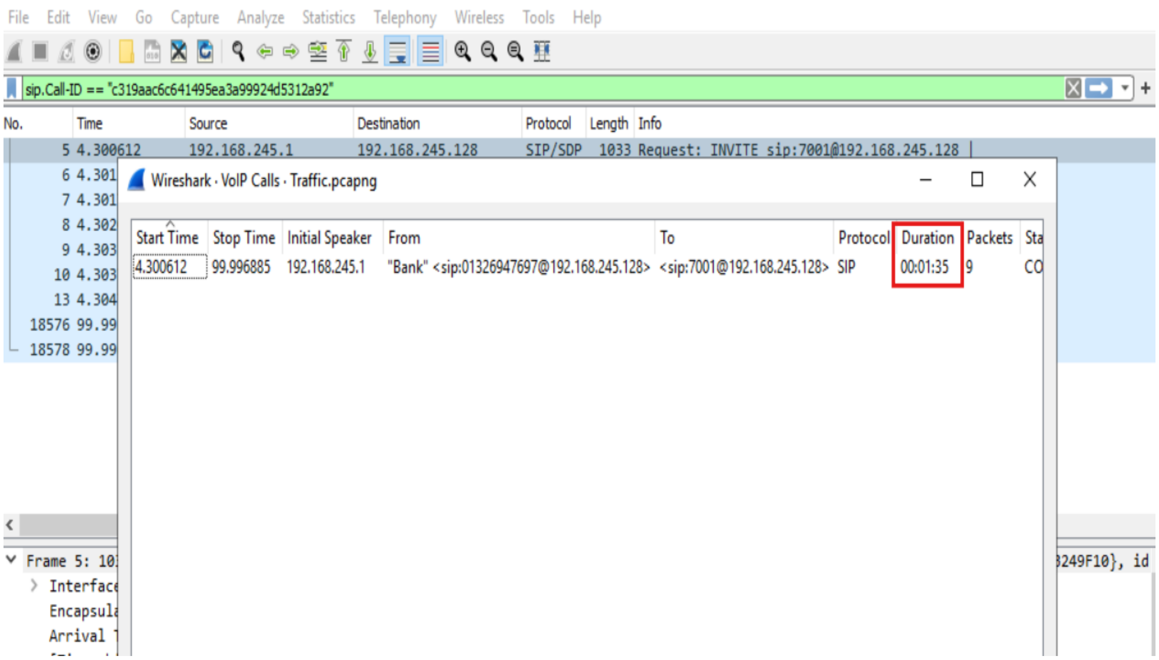


Based on our search in the first question, I checked the first displayed packet and identified the arrival time.

Q3: What is James's phone number?

I used the filter sip.Method == "INVITE" to find the James's phone number.

Q4: How long was the call with the bank?



For the fourth question, I used the "VoIP Calls" tool under the Telephony menu to automatically display the call details and read the exact duration from the "Duration" column.

Q5: What is the phone number of the bank that James received a call from?

```
Contact: "Bank" <sip:01326947697@192.168.245.1:59833;ob>
Call-ID: c319aac6c641495ea3a99924d5312a92
CSeq: 21391 INVITE
Allow: PRACK, INVITE, ACK, BYE, CANCEL, UPDATE, INFO, SUBSCRIBE, NOTIFY, REFER, MESSAGE,
OPTIONS
Supported: replaces, 100rel, timer, norefersub
Session-Expires: 1800
Min-SE: 90
User-Agent: MicroSIP/3.21.3
Content-Type: application/sdp
Content-Length:   335
```

For the fifth question, I inspected the SIP header fields, specifically looking at the "Contact" header, to identify the spoofed bank's phone number from the SIP URI.

Executive Summary

This investigation analyzed a network traffic capture to reconstruct a vishing (voice phishing) attack where an attacker spoofed their identity as "Bank" to target a user named James. By dissecting the **SIP** signaling layer and isolating **RTP** media payloads within Wireshark, the analysis successfully mapped out the entire incident without needing to listen to the audio directly. The investigation uncovered the critical details of the fraudulent encounter, showing that the attacker initiated the call from phone number **01326947697** (IP 192.168.245.1) to James's extension **7001** on **May 3, 2024, at 20:36:36 UTC**, with the entire deceptive conversation lasting exactly **00:01:35**